21st Karnataka State Level ISTE Student Convention, 2025

"Next-Gen Engineering: Innovating for India's Deep Tech Future"

Date: 11th and 12th November 2025
Venue: Yenepoya Institute of Technology, Moodbidri

<u>PROBLEM STATEMENTS- WEB/APP DEV</u>

1. Professional Networking and Career Development Platform

Objective:

Build a comprehensive professional networking platform that connects students, professionals, and organizations for career growth and knowledge sharing. The system should enable users to create detailed professional profiles with work experience, education, skills, projects, and certifications. Implement intelligent connection suggestions based on mutual networks, educational background, and professional interests with 1st, 2nd, and 3rd degree connection levels. Include a job portal with advanced search filters, one-click applications, and referral tracking. Features must include content sharing with posts, articles, and videos with engagement metrics, messaging system with file sharing, professional groups and communities, company pages with employee reviews, skill endorsements and recommendations from connections, course marketplace for professional development, and personalized feed algorithm showing relevant content from network.

Advanced features should include AI-powered career assistant, salary insights and negotiation tools, mentor-mentee matching, event management for professional meetups, analytics dashboard showing profile views and network growth, resume builder from profile data, job alerts matching user preferences, application tracking system, and search functionality across people, jobs, companies, and content. Ensure scalability for 100,000+ users, real-time messaging and notifications, recommendation algorithms for connections and jobs, mobile-responsive design, and robust privacy controls. The platform should serve as a complete professional identity and networking solution enabling career advancement through meaningful connections.

2. Advanced Online Examination and Assessment System

Objective:

Develop a secure, intelligent examination platform supporting multiple question types including MCQs, descriptive answers, coding problems, and handwritten answer evaluation using AI. The system must handle exam creation with question bank management, difficulty tagging, and random paper generation following blueprints. Implement browser lockdown preventing tab switching, webcam proctoring with periodic photo capture, screen recording for suspicious activity detection, and AI-powered behavior analysis flagging anomalies. For handwritten answers, faculty should upload model answers and the system must use OCR and NLP to evaluate student submissions, providing scores and feedback on accuracy, completeness, and presentation quality.

Core features include automated grading for objective questions, partial marking support, plagiarism detection for descriptive answers, manual grading interface with inline annotations, scheduled exams with countdown timers, result analytics showing score distribution and question-wise performance, revaluation request workflow, answer script viewing for students, and performance comparison with class averages. Advanced capabilities should include adaptive testing adjusting difficulty based on responses, question analytics showing discrimination index, AI-powered question generation from syllabus, practice test mode, mobile app support with offline mode, integration with learning management systems, grade book with weighted components, parent/guardian access, detailed student performance reports identifying weak areas, bulk exam scheduling, role-based access for administrators, teachers, and students, and comprehensive security measures including encryption, audit logs, and cheat detection. The platform should handle concurrent exams for 1000+ students with sub-second response times.

3. **Smart Payments in Healthcare SaaS. (by Adit (USA))**

Context:
In today's healthcare landscape, patients expect transparent, secure, and seamless payment experiences. Our SaaS platform integrates with multiple **Electronic Health Record (EHR)** systems to help hospitals and clinics manage billing, claims, and patient payments efficiently.
However, one of the critical challenges faced by healthcare providers is

managing **refunds** and **payment processing fees** transparently. Refunds can occur due to claim adjustments, service cancellations, or billing errors. At the same time, clinics often struggle to handle **processing charges** (e.g., credit card or payment gateway fees) deciding whether to absorb them or pass them to the patient.
Your challenge is to **design and prototype** a **smart payment module** that automates these processes while ensuring compliance, transparency, and a great user experience for both patients and administrators.

The Challenge:

Build a Refund and Payment Charges Management Feature within a SaaS Healthcare Payments Module that:

1. **Integrates seamlessly with EHR systems** to pull relevant patient billing data, payment records, and service details.
2. **Implements a secure and compliant refund process** that:
    a. Validates refund eligibility based on the original payment and service data.
    b. Allows partial or full refunds.
    c. Updates the EHR and financial systems automatically once a refund is processed.
    d. Sends notifications to patients and finance admins.
3. **Automatically applies payment processing charges** during payment collection by:
    a. Detecting the payment method (credit/debit/ACH, etc.).
    b. Calculating and appending the applicable processing fee dynamically.
    c. Displaying a transparent breakdown of fees before payment confirmation.
    d. Handling exceptions where certain fees are waived or absorbed by the provider.
4. **Ensure compliance** with healthcare data and payment regulations (HIPAA, PCI-DSS, etc.).
5. **Provides clear analytics and audit trails** for administrators tracking refunds, chargebacks, and processing fees.
6. **Optimizes patient experience**, offering transparency, simplicity, and trust.

Key Objectives:

- Create intuitive **UI/UX workflows** for refunds and charge management.

- Enable **real-time synchronization** between the SaaS platform and connected EHR systems.
- Ensure **data security and integrity** across all payment-related transactions.
- Deliver **actionable insights** to admins, e.g., refund trends, fee recovery rates, payment gateway costs.
- Support **scalability** for multiple healthcare facilities and payment gateways.

Expected Deliverables:

- A **working prototype** (web or mobile) demonstrating:
    - Refund request and approval workflow.
    - Real-time EHR integration for billing and service validation.
    - Automated payment processing fee calculation and display.
    - Transparent patient billing summary before and after transactions.

- A short **demo or presentation** highlighting:
    - The technical architecture (API design, integration with EHR, etc.).
    - Security and compliance measures.
    - User flow (for patients, admins, and finance teams).

Evaluation Criteria:

- **Innovation & Impact** (How well does your solution improve transparency and automation?
- **Integration Feasibility** (EHR and payment gateway interoperability)
- **User Experience** (Clarity, accessibility, ease of use for patients and staff)
- **Scalability & Security** (HIPAA/PCI compliance, multi-tenant architecture)
- **Demo Quality** (Clarity and completeness of the working solution)

Example Use Case:
A patient pays $200 for an outpatient service using a credit card.

- The platform automatically adds a $3.50 processing fee before checkout.
- Later, due to an overbilling correction, a $50 refund is initiated.
- The refund process validates the transaction via the EHR, initiates a refund through the payment gateway, and updates both the EHR and SaaS dashboards in real time.
- Both patient and admin receive notifications, and an audit log is created

for compliance.

Technical Problem Breakdown

1. System Components

Participants must design a modular architecture comprising:

Component	Description
Frontend (Portal or UI)	Interface for patients (payments/refunds) and admins (refund approvals, reports).
Payments Service	Core microservice handling payment initiation, processing charge computation, and gateway integration.
Refund Service	Handles refund requests, validations, workflows, and transaction reversals.
EHR Integration Layer	API middleware to fetch and update patient, billing, and encounter data from the connected EHR.
Database	Stores payment records, refunds, logs, and audit trails.
Notification Service	Sends alerts via email/SMS/app for successful payments, refunds, and errors.
Analytics Dashboard	Displays metrics like refund rates, processing fee collections, etc.

2. Key Functional Requirements

a. Refund Processing

- Validate refund eligibility by verifying:
  - Original payment exists in EHR and SaaS DB.
  - Refund amount ≤ original amount.
  - Refund reason (overpayment, service cancellation, claim adjustment, etc.).
- Handle **partial and full refunds**.

- Trigger refund via **payment gateway API**.
- Sync refund transaction with:
  - EHR (update billing and payment status).
  - SaaS database (for analytics and compliance).
- Generate **audit logs** and send **notifications** to patients and finance teams.

b. Payment Processing Charges

- Fetch **payment method type** (credit card, ACH, wallet, etc.).
- Apply **configurable fee rules**, e.g.:
  - Credit Card → +2.9% + $0.30
  - ACH → +1%
  - Internal Wallet → No Fee
- Display a **transparent fee breakdown** in the UI before patient confirms payment.
- Record the **fee component** separately in transaction metadata for accounting.

c. EHR Integration

- Use standardized healthcare interoperability formats (FHIR, HL7, or vendor APIs) to:
  - Fetch: Patient, Encounter, Invoice, PaymentNotice resources.
  - Update: PaymentReconciliation, Invoice with refund and payment status.
- Handle OAuth 2.0 or API key-based authentication for secure EHR access.

4. Comprehensive Learning Management System with Adaptive Intelligence

Objective:

Develop an enterprise-grade LMS delivering personalized, adaptive learning experiences beyond traditional content delivery. The platform should support modular course creation with weeks, units, and lessons, multimedia content upload including videos, PDFs, presentations, SCORM packages, and interactive simulations. Implement AI-powered adaptive learning that analyzes student performance and adjusts content difficulty, recommends supplementary materials, and creates personalized learning paths. Include prerequisite enforcement, branching scenarios based on quiz performance, and mastery-based progression where students must demonstrate competency before advancing.

Core features include interactive video player with in-video quizzes, note-taking with timestamps, playback speed control, live class integration with breakout rooms and whiteboard, discussion forums with threading and expert answers, assignment creation supporting multiple formats (text, file, code repository, presentation), peer review workflows with rubric-based evaluation, automated and manual grading with inline feedback, plagiarism detection, grade book with weighted categories and curve grading, progress dashboards showing completion percentage and time spent, learning analytics identifying struggling students early with intervention recommendations, gamification with badges and certificates, mobile-responsive design with offline content access. Advanced capabilities should include AI tutor chatbot for instant doubt resolution, spaced repetition algorithm for knowledge retention, microlearning modules for bite-sized content, collaborative group projects with shared workspaces, integration with video conferencing tools, calendar showing all deadlines, parent/guardian progress monitoring, accessibility features including screen reader support and closed captions, skill mastery tracking mapped to learning outcomes, competency-based assessments, social learning features encouraging peer collaboration, content effectiveness analysis showing correlation between materials and performance, A/B testing for instructional approaches, course templates for rapid deployment, external resource recommendations from educational platforms, and API integrations with student information systems. The system must support 10,000+ concurrent users, provide sub-second page loads, maintain SCORM compliance, ensure data privacy, and include comprehensive admin controls for user management, course approval, and platform analytics

5. Integrated College Management and Administration System

Objective:

Build an all-encompassing digital platform that transforms paper-based college operations into a unified, intelligent management system covering all administrative, academic, financial, and operational functions. The system should serve as a single source of truth connecting students, faculty, staff, parents, and management through role-based portals with seamless data flow across departments eliminating redundant manual processes.

Core Modules: The platform must integrate five essential modules working cohesively. **Human Resources Management** handles complete employee lifecycle including recruitment with job posting and application tracking, digital onboarding with document collection, attendance management with bulk upload capability, leave management with multi-level approval workflows, automated payroll processing with configurable salary structures including allowances, deductions, tax calculations, PF/ESI components, digital salary slip generation, increment and appraisal workflows with performance reviews, asset allocation tracking, training records, exit management with departmental clearances, and comprehensive HR analytics.

Admission and Accounts features online application portal with document upload, automated merit list generation and seat allocation, comprehensive fee management supporting multiple structures and installment plans, integrated payment gateway, scholarship and concession workflows, automated defaulter reminders, refund processing, complete accounting with income-expense tracking, vendor payments, bank reconciliation, budget monitoring by department, financial reporting with audit trails, and GST calculations.
Hostel Management covers digital room allocation with floor plans, online allotment based on preferences with roommate matching, visitor logging, gate pass system with parent notifications, mess management with menu display and feedback, mess bill calculation, maintenance complaint tracking with priority assignment, disciplinary records, and warden dashboard showing real-time occupancy and pending issues.
Timetable and Classroom Allotment provides intelligent scheduling with drag-and-drop interface, automatic conflict detection, constraint management for faculty availability, automated timetable generation with manual override capability, workload balancing reports, substitution management for absent faculty, classroom utilization analytics, exam scheduling with invigilator assignment, personalized schedule portals for students and faculty with PDF export and calendar sync, room booking system with approvals, and comprehensive analytics.
Transport Management includes route master with all stops and timings, vehicle and driver database, online bus pass application with digital QR code generation, transport fee tracking, maintenance scheduling, fuel consumption monitoring, trip sheets, complaint management, and utilization reports.

Additional Features: Implement an **AI Chatbot Assistant** supporting natural language queries in multiple languages, answering questions about admissions, fees, schedules, and college information with context awareness, real-time data integration from all modules, personalized responses by role, automatic ticket generation for complex queries, and analytics dashboard. Provide **role-specific**

dashboards for management showing institution-wide KPIs and pending approvals, HODs viewing department metrics and budget utilization, faculty accessing timetables and salary information, students viewing attendance and fee status, and parents monitoring their child's progress. Include **comprehensive reporting** with custom report builder supporting filtering, grouping, Excel/PDF export, scheduled reports, pre-built templates across modules, interactive visualizations, drill-down analytics, and complete audit logs tracking all system actions.

Advanced Capabilities: Support document management with digital storage, communication portal for targeted announcements, feedback and survey system, grievance redressal with escalation workflows, meeting scheduler, library integration, placement cell management, examination processing with hall tickets and results, digital certificate generation with verification, alumni database, event management, asset tracking, vendor management, and accreditation support.